

ANALYTICAL PROBLEM SOLVING – ASSESSMENT

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N K E V I N
M A T T H E W . Y

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Course / Subject: Artificial Intelligence

1. Water Jug Problem

Given: A 4-gallon jug and a 3-gallon jug are available. Both jugs have no measurement markings.

Goal: Measure exactly **2 gallons of water in the 4-gallon jug**.

State Representation:

A state is represented as (**amount of water in 4-gallon jug, amount of water in 3-gallon jug**).

Solution:

1. Fill the 3-gallon jug.
2. Pour it into the 4-gallon jug.
3. Fill the 3-gallon jug again.
4. Pour it into the 4-gallon jug until the 4-gallon jug becomes full.
5. The 3-gallon jug will have 2 gallons remaining.
6. Empty the 4-gallon jug.
7. Transfer the remaining 2 gallons into the 4-gallon jug.

Result: Exactly **2 gallons of water** are obtained in the 4-gallon jug.

2. Mars Rover – Intelligent Agent

i. Percepts:

The rover observes rocks, soil, obstacles, temperature, atmosphere, distance, terrain, sample sites, and possible signs of water or minerals.

ii. Operating Environment:

The Mars rover works in an environment that is **partially observable, dynamic, sequential, uncertain, continuous, and mostly single-agent**.

iii. Actions:

The rover can move, rotate, collect and examine samples, capture images, avoid obstacles, navigate different areas, and send information to Earth.

iv. Performance Measures:

Its performance can be evaluated based on the number of useful samples collected, accuracy of analysis, distance covered, time used, energy consumption, safe obstacle avoidance, scientific discoveries, and successful completion of the mission.

v. Suitable Architecture:

A **Model-Based Utility-Based Agent** is suitable. It keeps track of information about its surroundings and selects actions by considering useful goals such as collecting samples, maintaining safety, and saving energy.

3. 8-Queens Problem

Problem:

The objective is to arrange **8 queens on an 8×8 chessboard** so that no queen can attack another queen.

Initial State:

The chessboard is completely empty.

State:

A state represents a partial placement of queens on the board.

Actions:

Place the next queen in a position where it cannot be attacked by any previously placed queen.

Transition Model:

Placing a queen in a safe position creates a new valid state.

Goal Test:

The solution is reached when all 8 queens are placed without any two queens sharing the same row, column, or diagonal.

Path Cost:

Each queen placement is assigned a cost of 1. Therefore, placing all 8 queens gives a total cost of **8**.

Suitable Search:

Backtracking / Depth First Search (DFS) is suitable because an incorrect arrangement can be rejected immediately.

One Valid Solution:

[1, 5, 8, 6, 3, 7, 2, 4]

Here, each number indicates the row position of the queen in each column from left to right.

4. OLA Cab Booking – Problem-Solving Agent

Problem:

A passenger needs to travel from a starting location to a destination using an OLA cab service. The passenger can choose options such as **Mini, Micro, Sedan, Prime, or Shared**, depending on price, comfort, and travel needs.

Agent Type:

Utility-Based Agent

Reason:

The agent evaluates different cab options by considering parameters such as **fare, journey time, comfort, and customer preferences**. It then chooses the option that provides the greatest overall utility.

Pseudocode

START

1. Enter the source location.
2. Enter the destination location.
3. Search for available cabs.
4. For every available cab:
 1. Find the estimated fare.
 2. Calculate the expected travel time.
 3. Check the comfort level.
 4. Determine the utility value.
5. Choose the cab with the highest utility.
6. Show the selected cab, fare, and estimated travel time.
7. Ask the customer for confirmation.
8. If the customer accepts:
 1. Confirm the cab booking.
 2. Display the booking information.
9. Otherwise:
 1. Cancel the booking.

STOP

5. Logistics Delivery Network – Uniform Cost Search

Given Edge Costs:

- $S \rightarrow A = 1$
- $S \rightarrow G = 12$
- $A \rightarrow B = 3$
- $A \rightarrow C = 1$
- $B \rightarrow D = 3$
- $C \rightarrow D = 1$
- $C \rightarrow G = 2$
- $D \rightarrow G = 3$

Uniform Cost Search (UCS):

UCS always chooses and expands the path with the **smallest accumulated cost**.

Stage	Path / Node	Total Cost
Start	S	0
1	$S \rightarrow A$	1
2	$S \rightarrow A \rightarrow C$	2
3	$S \rightarrow A \rightarrow C \rightarrow D$	3
4	$S \rightarrow A \rightarrow C \rightarrow G$	4

Comparison:

The route $S \rightarrow A \rightarrow C \rightarrow G$ has a total cost of:

$$1 + 1 + 2 = 4$$

The direct route $S \rightarrow G$ has a cost of **12**.

Therefore, UCS chooses the route with the lower total cost.

Final Answer:

Least-cost path = $S \rightarrow A \rightarrow C \rightarrow G$

Minimum cost = 4